NETWORK RECONNAISSANCE REPORT

Metasploitable 2 Virtual Lab

Report detail	Value
Assessment type	Network reconnaissance and service enumeration
Testing machine	Kali Linux
Target machine	Metasploitable 2 (Linux)
Kali IP address	192.168.56.102
Target IP address	192.168.56.103
Assessment date	3 November 2026
Prepared by	Krish Sheemar

Scope: discovery, TCP port scanning, service identification and FTP enumeration. No exploitation or privilege escalation is claimed.

Executive Summary

A network reconnaissance assessment was performed in a VirtualBox laboratory using Kali Linux to examine a Metasploitable 2 target at 192.168.56.103. The supplied evidence shows that the target was reachable, multiple TCP services were exposed, and the FTP service identified itself as vsFTPd 2.3.4. Anonymous FTP authentication was accepted, and the FTP client returned a listing of the remote root directory. These observations identify services and configurations that merit security review.

The work documented here is limited to reconnaissance and service enumeration. The screenshots do not demonstrate successful remote command execution, exploitation, or root access.

1. Assessment Objectives

- Confirm that the target host is reachable from the Kali testing machine.
- Discover exposed TCP ports on the target.
- Identify network services and reported software versions.
- Check whether anonymous FTP authentication is permitted.
- Record evidence and provide proportionate security recommendations.

2. Lab Environment and Scope

Component	Observed configuration
Testing system	Kali Linux
Kali interface / address	eth1 / 192.168.56.102 (from earlier route output in the conversation)
Target system	Metasploitable 2
Target interface / address	eth0 / 192.168.56.103 (shown in Screenshot 1)
Virtualisation	Oracle VirtualBox
Assessment scope	Host discovery, TCP port scanning, service/version detection and FTP enumeration

The assessment is limited to the lab target identified above. No destructive testing, file modification, privilege escalation, or successful exploitation is evidenced in the supplied screenshots.

3. Methodology and Evidence

3.1 Target Network Configuration

The target VM's ifconfig output shows interface eth0 configured with IPv4 address 192.168.56.103 and netmask 255.255.255.0. This establishes the target address used for the subsequent scans.

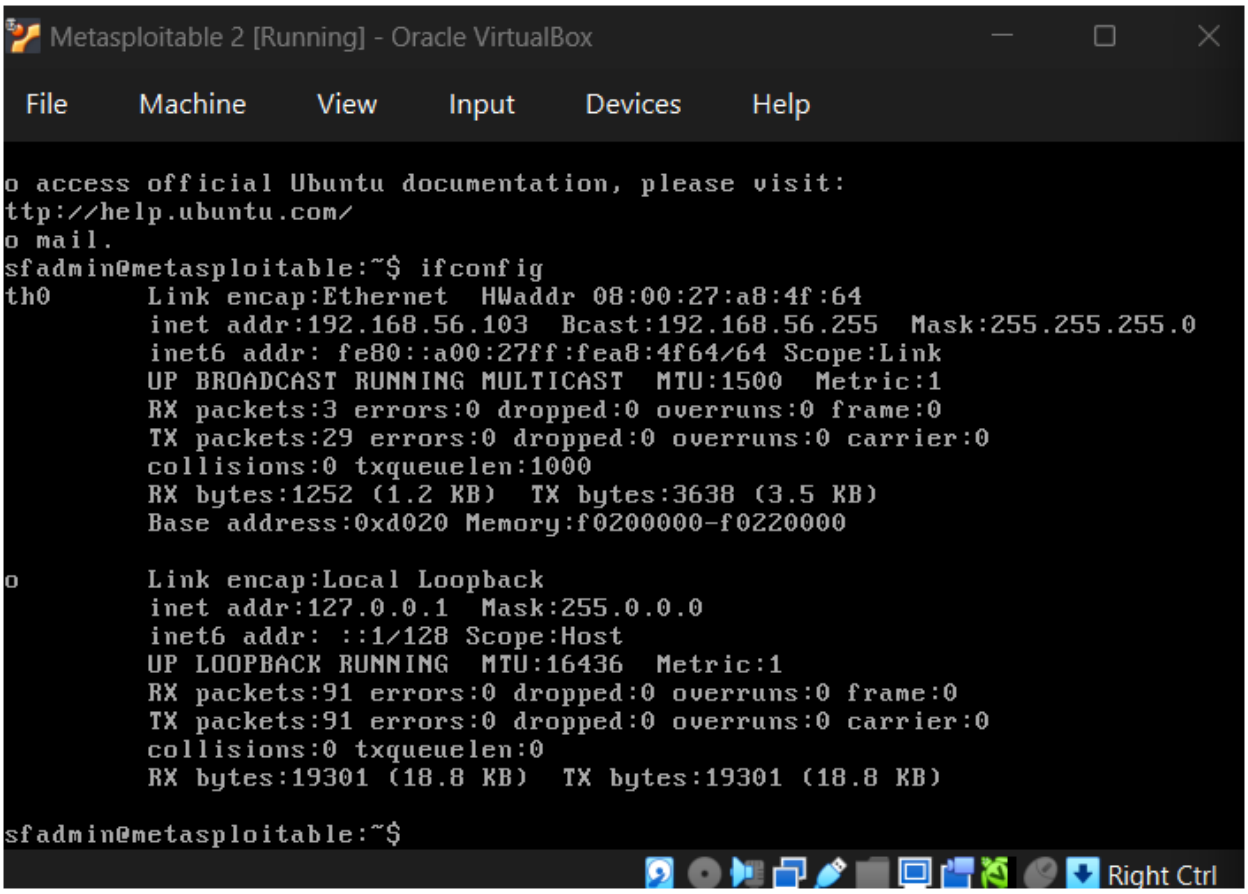


Figure 1. Metasploitable 2 target interface configuration (ifconfig).

3.2 Service and Version Detection

The Nmap service scan reported the host as up and identified the following open TCP services and version information. Nmap service labels and version strings are observations from the scan; they should be treated as service fingerprints rather than proof that every service is correctly configured.

```
nmap -sV 192.168.56.103
```

Port	Service	Nmap-reported version / detail
21/tcp	FTP	vsftpd 2.3.4
22/tcp	SSH	OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp	Telnet	Linux telnetd
25/tcp	SMTP	Postfix smtpd
53/tcp	DNS	ISC BIND 9.4.2
80/tcp	HTTP	Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp	RPCBind	2 (RPC #100000)
139/tcp	NetBIOS/SMB	Samba smbd 3.X–4.X (WORKGROUP)
445/tcp	SMB	Samba smbd 3.X–4.X (WORKGROUP)
512/tcp	exec	netkit-rsh rexecd
513/tcp	login	OpenBSD or Solaris rlogind
514/tcp	shell	Netkit rshd

1099/tcp	Java RMI	GNU Classpath grmiregistry
1524/tcp	bindshell	Metasploitable root shell (Nmap service label)
2049/tcp	NFS	2–4 (RPC #100003)
2121/tcp	FTP	ProFTPD 1.3.1
3306/tcp	MySQL	MySQL 5.0.51a-3ubuntu5
5432/tcp	PostgreSQL	PostgreSQL DB 8.3.0–8.3.7
5900/tcp	VNC	VNC (protocol 3.3)
6000/tcp	X11	Access denied
6667/tcp	IRC	UnrealIRCd
8009/tcp	AJP13	Apache Jserv (Protocol v1.3)
8180/tcp	HTTP	Apache Tomcat/Coyote JSP engine 1.1

The supplied service scan screenshot reports 977 closed TCP ports and lists the open services shown above. The separate full-port scan screenshot reports approximately 30 open ports; the service scan screenshot displays the detailed list above. Differences between scan screenshots can occur because the commands, scan timing, or displayed output may differ. The final submitted scan output should be used as the authoritative record.

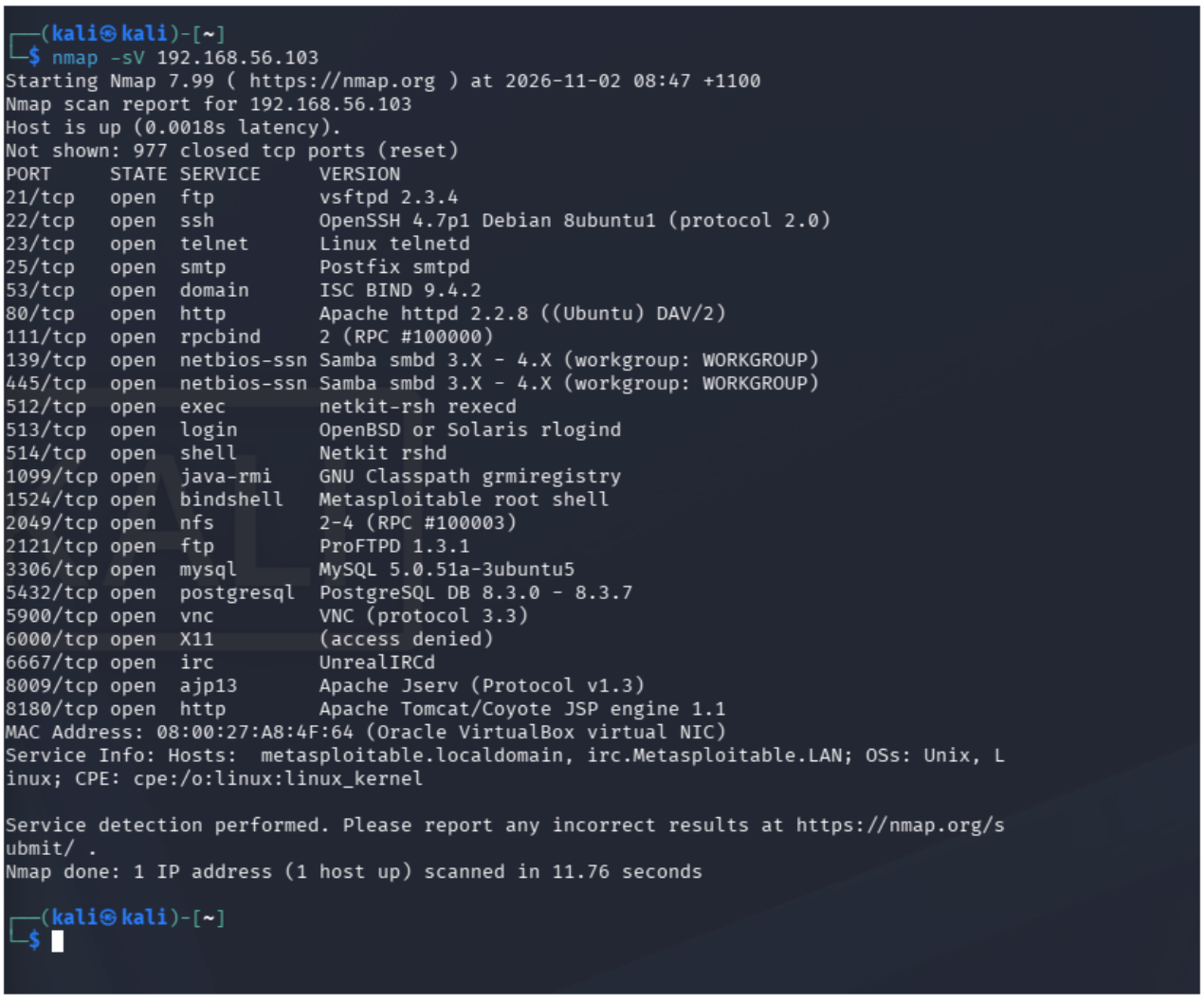


Figure 2. Nmap service and version detection results for 192.168.56.103.

3.3 Focused FTP, SSH and Telnet Enumeration

A focused Nmap scan was run against ports 21, 22 and 23 using default scripts and service detection. The FTP service was identified as vsFTPd 2.3.4. The ftp-syst script reported that the control and data connections use plain text. The ftp-anon script reported that anonymous FTP login was allowed (FTP code 230). SSH host-key fingerprints were also collected, and Telnet was identified as Linux telnetd.

```
nmap -sC -sV -p 21,22,23 192.168.56.103
```

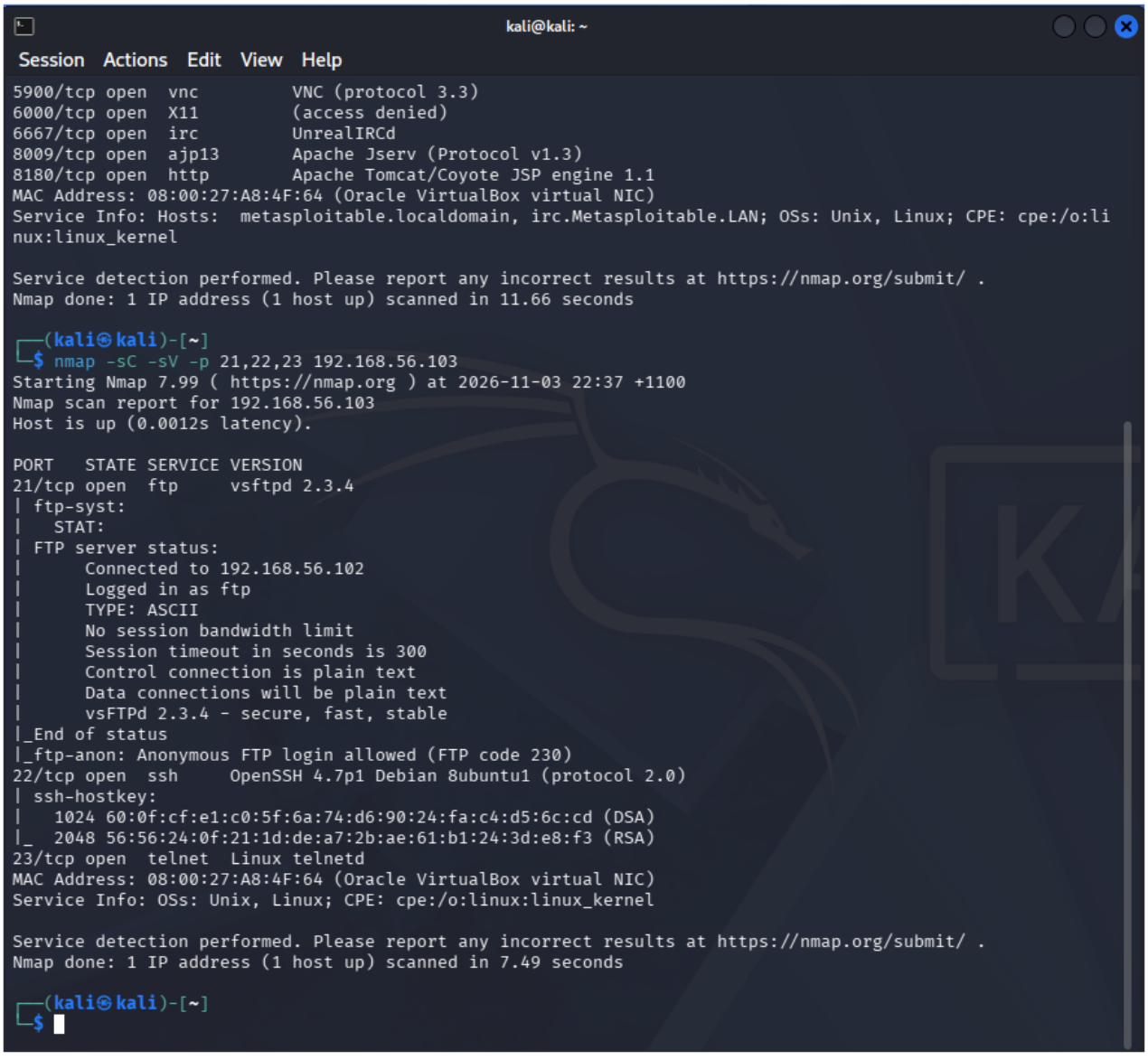


Figure 3. Focused Nmap enumeration of FTP, SSH and Telnet.

3.4 Anonymous FTP Check

The ftp-anon Nmap script independently reported that anonymous FTP login was allowed. This confirms that the FTP service accepted an anonymous login during the test.

```
nmap -p 21 --script ftp-anon 192.168.56.103
```

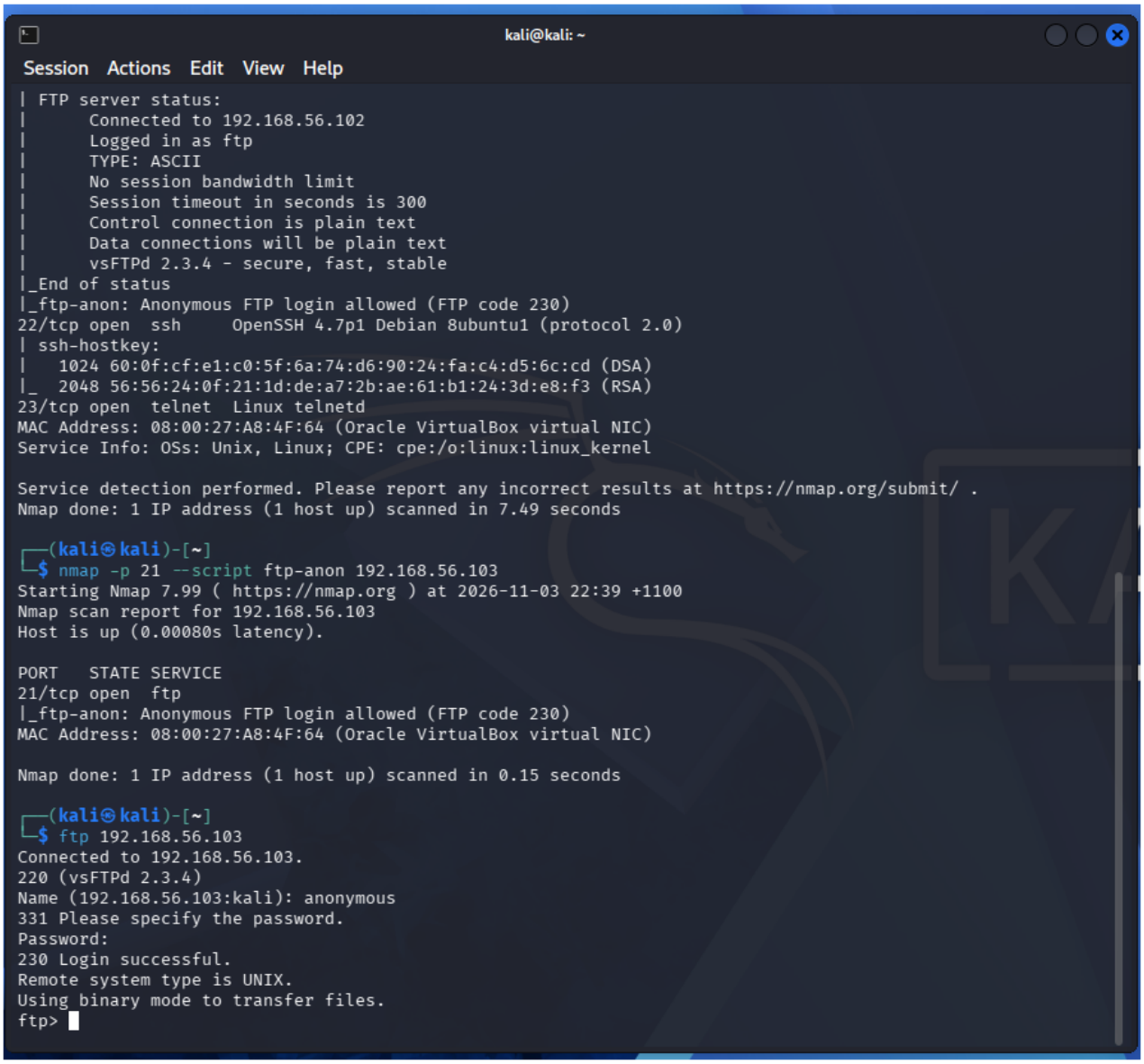


Figure 4. Nmap ftp-anon script result showing anonymous FTP login allowed.

3.5 Manual FTP Connection and Directory Listing

A manual FTP client connection was established to 192.168.56.103. The server banner identified vsFTPd 2.3.4. The username anonymous was accepted, and the server returned FTP response code 230 (Login successful). The pwd command returned the remote directory '/'. The ls -la command returned a directory listing containing the current and parent directory entries.

```
ftp 192.168.56.103
Name: anonymous
ftp> pwd
ftp> ls -la
ftp> bye
```

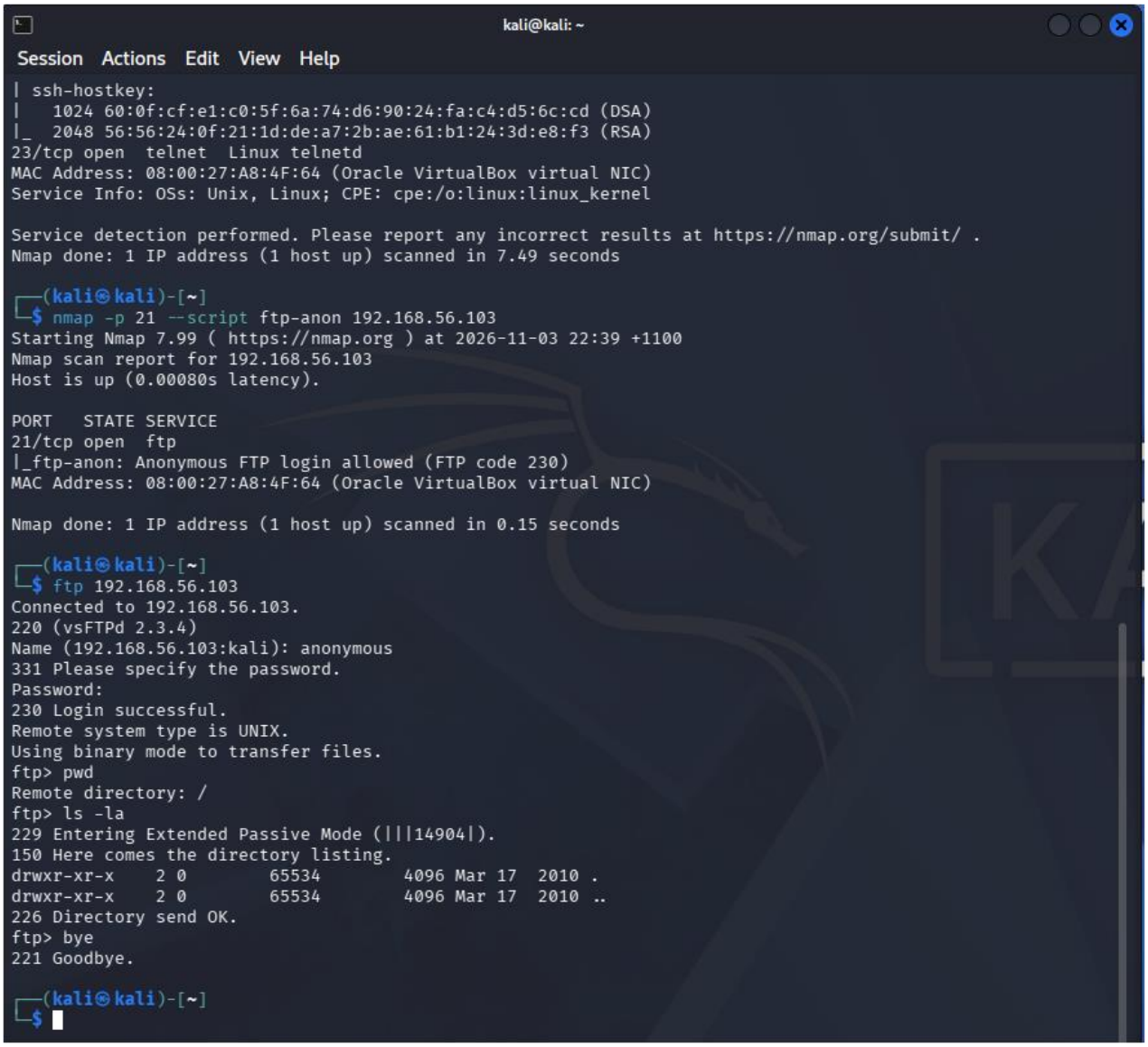


Figure 5. Anonymous FTP login, remote working directory and directory listing.

4. Key Findings and Security Implications

Finding	Evidence / implication	Recommendation
F1 – Multiple exposed TCP services	The target exposes numerous services, including remote access, file sharing, databases, web and RPC services. Each listening service increases the attack surface.	Review whether each service is required; restrict access using host and network firewalls.
F2 – Anonymous FTP enabled	Anonymous authentication was accepted on TCP/21. The shown listing did not reveal files beyond '.' and '..', but access was still granted.	Disable anonymous FTP unless explicitly required. If required, restrict it to a dedicated read-only directory.
F3 – FTP uses plaintext	The FTP status script reported that control and data connections are plain text.	Replace FTP with SFTP or FTPS where appropriate, and avoid sending credentials over untrusted networks.
F4 – Telnet exposed	TCP/23 was identified as Linux telnetd. Traditional Telnet does not encrypt session traffic.	Disable Telnet and use SSH for remote administration.

F5 – Legacy software fingerprints	The scan identified several older software versions, including vsFTPd 2.3.4, OpenSSH 4.7p1, Apache 2.2.8, MySQL 5.0.51a and ProFTPD 1.3.1.	Confirm exact package versions and patch status, then upgrade or isolate unsupported services.

Note: Nmap's service label on TCP/1524 is 'Metasploitable root shell'. This is a port/service fingerprint reported by Nmap, not evidence that the assessor established a shell or authenticated to it. It should be reported as a service identification requiring controlled follow-up verification, not as a confirmed successful compromise.

5. Limitations

- The assessment was limited to the scans and FTP checks represented in the supplied screenshots.
- The screenshots do not include a complete saved XML or normal-format scan file for independent review.
- A service/version scan identifies likely services and versions but does not, by itself, prove exploitability.
- No successful remote command execution, exploitation, privilege escalation, or root access is demonstrated.
- The FTP directory listing shown contains only the current and parent directory entries; no accessible files are evidenced.

6. Recommendations

1. Disable anonymous FTP or confine it to a clearly defined read-only area.
2. Replace plaintext FTP with SFTP or FTPS and disable Telnet in favour of SSH.
3. Update unsupported or outdated services to vendor-supported versions.
4. Restrict database, SMB, NFS, RPC and remote administration ports to trusted systems that require them.
5. Apply host-based firewall rules and segment laboratory services from other networks.
6. Repeat service discovery after remediation and compare the results to verify that unnecessary exposure has been reduced.

7. Conclusion

The reconnaissance successfully identified the Metasploitable 2 target at 192.168.56.103 and documented a broad set of exposed TCP services. Service detection identified multiple legacy service fingerprints. FTP enumeration confirmed that anonymous authentication was accepted by vsFTPd 2.3.4, and a manual FTP session confirmed access to the remote root directory listing. The results provide a useful baseline of the target's network exposure and highlight opportunities to reduce risk through service hardening, encryption, patching and access restrictions.

Appendix A – Screenshot Index

Figure	Evidence
Figure 1	Target VM network interface configuration (ifconfig).
Figure 2	Nmap service and version detection.
Figure 3	Focused Nmap enumeration of FTP, SSH and Telnet.
Figure 4	Nmap anonymous FTP script result.
Figure 5	Manual anonymous FTP login and directory listing.

Before submission: Insert your name and student details, verify that the service-scan output is the final version you intend to submit, and follow your instructor's required report format.